Block practice · Research methods

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Calculating the mean

Spotting the IV and the DV

Arithmetic: standard form, estimates and rounding

Extraneous variables and how to control them

Experiments: lab, field and natural

OLevels.com SCHOOL

Practice workbook · Calculating the mean

Finding or describing the arithmetic mean of a data set.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-15

1. Question 2 · part c · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

2 Jenna investigated whether stress affected performance on a memory test.

She recruited 10 participants from her university. Participants were asked to remember a list of 20 words. Fifteen minutes later, they were asked to write down as many words as they could remember. The same participants were tested twice:

- Test 1: one week before their examinations began.
- Test 2: one week after their examinations had finished.

(a) Describe how Jenna could use an opportunity sampling method to gather the participants for her investigation.

(2)

(b) State which experimental research design Jenna used in her investigation.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

1. Question 2 · part c · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Jenna recorded the total number of correct words remembered by participants both before and after their examinations. Her results are shown in **Table 2**.

Participant	Total number of correct words remembered one week before examinations began (out of 20)	Total number of correct words remembered one week after examinations finished (out of 20)
A	5	8
B	6	7
C	7	9
D	4	8
E	5	7
F	3	7
G	9	9
H	10	11
I	6	6
J	5	8

Table 2

OLevels.COM SCHOOL

(c) Calculate the mean score for the total number of correct words remembered one week **before** the participants' examinations began.

(1)

Mean .....

(d) Calculate the median score for the total number of correct words remembered one week **after** the participants' examinations finished.

(2)

Median .....

(Total for Question 2 = 6 marks)

1. Question 2 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part a · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

2 Michelle conducted a study to find out if 15 year olds had better long-term memory recall than 55 year olds. She gave two groups of participants one hour to memorise a set of 40 pictures. After two weeks, she asked the participants to recall as many of the pictures as they could.

Her results are shown in **Table 1**.

15 year olds	Number of pictures recalled correctly	55 year olds	Number of pictures recalled correctly
Participant A	35	Participant A	16
Participant B	30	Participant B	28
Participant C	29	Participant C	27
Participant D	18	Participant D	34
Participant E	38	Participant E	35
Participant F	31	Participant F	26
Participant G	24	Participant G	21

Table 1

(a) Calculate the mean score for the number of pictures recalled by 55 year olds.

You must give your answer to two decimal places.

(2)

Mean score for 55 year olds .....

(b) Calculate the median score for the number of pictures recalled by 15 year olds.

(2)

Median score for 15 year olds .....

(c) Calculate the range for the number of pictures recalled by all participants.

(1)

Range for all participants .....

(Total for Question 2 = 5 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.

3 Ryan used a questionnaire to assess whether students had a fixed or growth mindset in four secondary schools. He asked the students to complete the questionnaire online.

His results are shown in **Table 2**.

School	Number of students with a fixed mindset	Number of students with a growth mindset
A	458	931
B	211	439
C	641	838
D	189	271

Table 2

(a) Calculate the mean number of students with a fixed mindset.

(1)

Mean .....

(b) Calculate the total number of students who completed the questionnaire.

(1)

Total .....

(c) Calculate the range for the number of students with a growth mindset.

(1)

Range .....

(Total for Question 3 = 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

3. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part d · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part d only.

3 Quentin is investigating the capacity of short-term memory. He showed 20 objects to 10 participants for 30 seconds. Quentin then asked them to recall as many as they could.

His results are shown in **Table 2**.

Participant	Number of objects recalled
A	10
B	9
C	8
D	8
E	6
F	7
G	8
H	9
I	10
J	11

Table 2

(a) Calculate how many participants recalled nine objects as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 3 8 3 A 0 8 4 8

4. Question 3 · part d · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Calculate the median for the number of objects recalled.

(2)

Median .....

(c) Calculate the mode for the number of objects recalled.

(1)

OLevels.COM — SCHOOL —

Mode .....

(d) State how Quentin would calculate the mean score for the number of objects recalled.

(1)

.....

.....

.....

(Total for Question 3 = 5 marks)

4. Question 3 · part d · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Spotting the IV and the DV

Recognising which factor a study changes or compares (IV) and which outcome it measures (DV).

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-01

1. Question 1 · part a · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part a only.

1 Thomas recruited participants to investigate obedience to authority. The participants were asked to attend a university.

They were allocated to one of two conditions:

- Condition A: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in a uniform.
- Condition B: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in jeans and a t-shirt.

Thomas recorded how many participants obeyed the instruction to remain silent.

(a) Identify the dependent variable (DV) in Thomas' investigation.

(1)

- ☒ **A** Whether the man was wearing a uniform or jeans and a t-shirt.
- ☒ **B** The amount of time participants were given to answer the questions.
- ☒ **C** The number of participants who obeyed the man's instruction to remain silent.
- ☒ **D** The number of general knowledge questions participants were instructed to answer.

(b) Describe **one** participant variable that Thomas may have considered when conducting his investigation.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 1 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

2 Winston investigated gender differences in pro-social and anti-social behaviour. He placed a mobile phone on a seat near the local police station and recorded the responses of male and female passers-by to the mobile phone.

(a) Name the independent variable (IV) for Winston's investigation.

(1)

.....

.....

.....

(b) The results for Winston's investigation are shown in **Table 2**.

	Number of male passers-by	Number of female passers-by
Took the mobile phone to the police station	3	7
Walked away with the mobile phone	8	8
Walked past the mobile phone	7	15

Table 2

Calculate how many participants were male as a fraction of all participants.

You must give your answer in the lowest form.

(2)

Males as a fraction of all participants .....

(c) Winston claims that females are more pro-social than males.

Give **one** way that the data in **Table 2** could support Winston's claim.

(1)

.....

.....

.....

(Total for Question 2 = 4 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.

5 Cathy researched whether short-term memory improved when grouping was used for learning telephone numbers. She showed the participants 10 telephone numbers.

- Condition A learned whole telephone numbers, for example 01234567890
- Condition B learned grouped telephone numbers, for example 0123 4567 890

Each telephone number was shown for 15 seconds. Then, after a 30-second delay, participants were asked to write the telephone number down.

The results of Cathy's experiment are shown in **Table 3**.

	Number of correctly recalled telephone numbers	Number of incorrectly recalled telephone numbers
Condition A	3	7
Condition B	6	4

Table 3

(a) Identify the independent variable for Cathy's experiment.

(1)

.....

.....

(b) Cathy concluded that grouping improves recall in short-term memory.

Give **one** way that the data in **Table 3** could support this conclusion.

(1)

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8

P 6 6 5 7 5 R A 0 8 4 8

3. Question 5 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part a · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Gary conducted an experiment to test whether there was a difference in the ability of boys and girls to conserve number. He sampled 50 boys and 50 girls aged six years old.

Gary showed them the image in **Figure 1**.

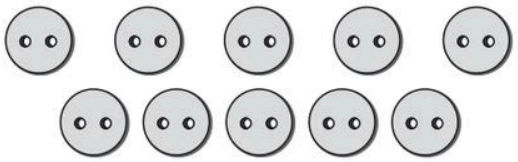


Figure 1

Gary asked each child to say whether the top row of buttons contained more, less, or the same number of buttons as the bottom row.

(a) Identify the independent variable in Gary's experiment.

(1)

- ☒ **A** age
- ☒ **B** gender
- ☒ **C** ability
- ☒ **D** conservation

OLevels.com SCHOOL

(b) Give **two** ways that Gary could ensure his experiment is ethical.

(2)

1 .....

.....

.....

2 .....

.....

.....

4. Question 1 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Arithmetic: standard form, estimates and rounding

Totals, differences, estimates, standard form, decimal places and significant figures.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-19

1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

OLevels.COM SCHOOL

1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 5 5 2 4 A 0 1 0 4 8

1. Question 5 · part f · 1 mark

2024 JUN · Paper 02

Worked-example retry · Attempt part f only.

DO NOT WRITE IN THIS AREA

(e) Calculate the median score for the units of alcohol consumed by the patient using **Table 3**.

(2)

Median .....

(f) On one day in March, the patient drank a total of 12.81924 units of alcohol.

Give the units of alcohol to **two** decimal places.

(1)

.....

(Total for Question 5 = 8 marks)

OLevels.COM SCHOOL

Working and answer

2. Question 3 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 Vanessa investigated age differences in driving speed. She conducted an observation by a road in the local town. Vanessa recorded the speed that young drivers and mature drivers drove their cars in miles per hour (mph).

The number of young drivers and mature drivers who were recorded driving at each speed is shown in **Table 3**.

	1mph to 10mph	11mph to 20mph	21mph to 30mph	31mph to 40mph
Young driver	1	4	9	12
Mature driver	1	5	14	18

Table 3

(a) Calculate the difference between the number of young drivers who were recorded driving at 1mph to 20mph and the number of young drivers who were recorded driving at 21mph to 30mph.

(1)

Difference .....

(b) Calculate the ratio of young drivers to mature drivers that Vanessa recorded driving at 31mph to 40mph.

(1)

Ratio .....

(c) Explain **one** improvement that Vanessa could make to her investigation.

(2)

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 4 marks)

2. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

(1)

Number of families with children who completed the online questionnaire .....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10

P 5 9 6 6 9 A 0 1 0 4 8

3. Question 6 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 6 · part b · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b only.

6 Kayla investigated whether the use of positive facial expressions (such as smiles) by actors in television advertisements resulted in people being more likely to buy the product being advertised.

First, she analysed two newspaper articles about the use of facial expressions in television advertisements.

Kayla then used an online questionnaire to find out if facial expressions influenced the purchase of a particular brand of cereal. 39,980,000 responses were collected in total.

(a) Give 39,980,000 in standard form.

(1)

39,980,000 in standard form .....

(b) 10,003,010 responses showed positive facial expressions used by actors in television advertisements did not influence their decision to buy the cereal.

Estimate the number of responses that showed positive facial expressions **did** influence their decision to buy the cereal.

(1)

Estimation .....

(c) 63% of the total responses to the online questionnaire were completed by families with children.

Calculate the number of families with children who completed the online questionnaire.

(1)

Number of families with children who completed the online questionnaire .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

10

P 5 9 6 6 9 A 0 1 0 4 8

4. Question 6 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part e · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

OLevels.COM SCHOOL

(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

5. Question 3 · part e · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

(e) Becca's results are shown in **Table 2**.

	Mean time taken to accurately complete the coordination task (seconds)	Mean time taken to accurately complete the word-meaning task (seconds)
Males	57.24	63.51
Females	64.43	51.36

Table 2

Calculate the difference between the mean time (seconds) taken by **males** to accurately complete the coordination task and word-meaning task.

You must give your answer to two significant figures.

(2)

OLevels.COM — SCHOOL —

Difference in mean time (seconds) taken by males .....

(f) Explain **one** conclusion that Becca could make from her results.

(2)

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 10 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN

5. Question 3 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part b · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part b only.

3 Ryan used a questionnaire to assess whether students had a fixed or growth mindset in four secondary schools. He asked the students to complete the questionnaire online.

His results are shown in **Table 2**.

School	Number of students with a fixed mindset	Number of students with a growth mindset
A	458	931
B	211	439
C	641	838
D	189	271

Table 2

(a) Calculate the mean number of students with a fixed mindset.

(1)

Mean .....

(b) Calculate the total number of students who completed the questionnaire.

(1)

Total .....

(c) Calculate the range for the number of students with a growth mindset.

(1)

Range .....

(Total for Question 3 = 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

6. Question 3 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 5 · part c · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

OLevels.COM SCHOOL

7. Question 5 · part c · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part c only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 5 5 2 4 A 0 1 0 4 8

7. Question 5 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Extraneous variables and how to control them

Participant and situational variables, ways of controlling them (standardisation, blind techniques) and the choice of experimental design.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-02

1. Question 1 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

1 Thomas recruited participants to investigate obedience to authority. The participants were asked to attend a university.

They were allocated to one of two conditions:

- Condition A: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in a uniform.
- Condition B: 30 participants were instructed to answer general knowledge questions in silence for 40 minutes by a man dressed in jeans and a t-shirt.

Thomas recorded how many participants obeyed the instruction to remain silent.

(a) Identify the dependent variable (DV) in Thomas' investigation.

(1)

- ☒ **A** Whether the man was wearing a uniform or jeans and a t-shirt.
- ☒ **B** The amount of time participants were given to answer the questions.
- ☒ **C** The number of participants who obeyed the man's instruction to remain silent.
- ☒ **D** The number of general knowledge questions participants were instructed to answer.

(b) Describe **one** participant variable that Thomas may have considered when conducting his investigation.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 1 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

The results of Thomas' investigation are shown in **Table 1**.

	Obeyed instruction (remained silent)	Disobeyed instruction (did not remain silent)
Condition A: Man wearing uniform	22	8
Condition B: Man wearing jeans and t-shirt	11	19

Table 1

(c) Calculate the percentage of participants who **obeyed** the instruction, out of all the participants in the investigation.

(2)

Percentage .....

(d) Calculate the ratio of participants who **disobeyed** the instruction, to all participants taking part in the investigation.

(2)

Ratio .....

(e) Thomas concludes that the clothes worn by an authority figure influence levels of obedience.

Give **one** way Thomas' results could support this conclusion.

(1)

.....

.....

.....

(Total for Question 1 = 8 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

1. Question 1 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part c · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

OLevels.COM SCHOOL

(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

2. Question 3 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4 Ricky used a field experiment to investigate whether there is a difference between English and Sociology students' obedience to an authority figure.

He arranged for an English teacher and Sociology teacher to give their students an instruction to complete a piece of work. The teachers then left the classroom for 45 minutes.

Ricky recorded the number of students who did and did not complete the work. The results are shown in **Table 3**.

	Number of students who did complete the work	Number of students who did not complete the work
English students	16	8
Sociology students	15	10

Table 3

(a) Calculate the percentage of students who completed the work, out of all the students in the experiment.

(2)

OLevels.COM — SCHOOL —

Percentage of students who completed the work .....

3. Question 4 · part d · 1 mark

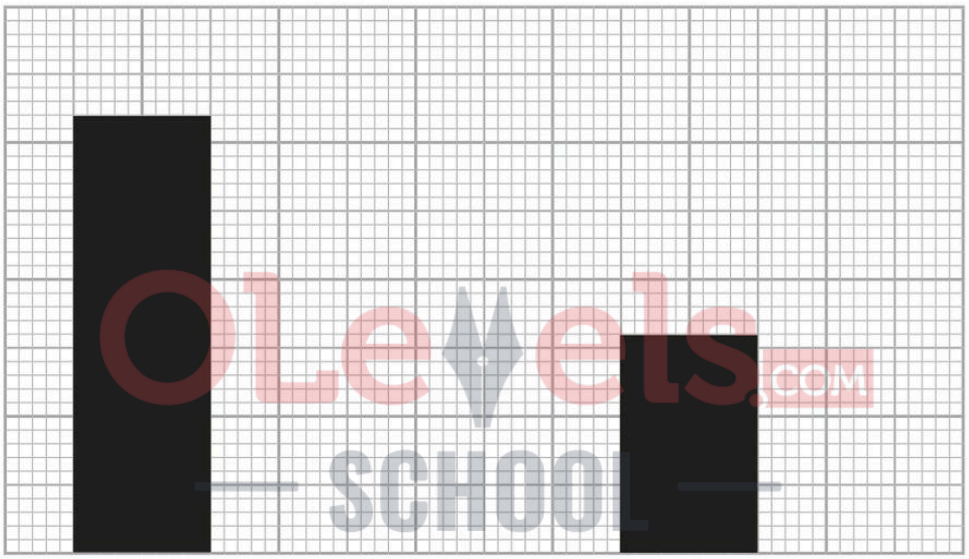
2020 NOV · Paper 02

Further practice · Attempt part d only.

(b) Complete the bar chart with the data from **Table 3** for the Sociology students, including the label and scale for the y-axis.

(3)

A bar chart to show the number of English and Sociology students who did and did not complete the work set by the teacher.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 4 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

(c) Ricky concluded that there was no significant difference between English and Sociology students' obedience to an authority figure.

Give **one** way that Ricky's findings could support this conclusion.

(1)

.....

.....

.....

(d) Give **one** participant variable that Ricky could have controlled for in his experiment.

(1)

.....

.....

.....

(Total for Question 4 = 7 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

Working and answer

4. Question 1 · part b · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Andrew conducted a laboratory experiment to investigate the capacity of short-term memory. He showed participants 20 trigrams and asked them to recall as many of the trigrams as they could.

(a) Andrew counted how many trigrams each participant recalled correctly.

Identify the type of data gathered by Andrew.

(1)

☒ **A** Qualitative data

☒ **B** Subjective data

☒ **C** Quantitative data

☒ **D** Secondary data

(b) Andrew used a standardised procedure when showing the trigrams to his participants.

Give **two** ways that Andrew could have standardised his procedure.

(2)

1 .....

.....

.....

2 .....

.....

.....

OLevels.com SCHOOL

4. Question 1 · part b · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

(c) Andrew sampled a total of 20 participants. A quarter of the participants were male.

Calculate the ratio of males to females in the participant sample used by Andrew in his experiment.

(2)

Ratio .....

(d) The number of trigrams recalled by participants correctly in Andrew's experiment are shown in **Table 1**.

	Number of trigrams recalled correctly				
	Between 1 and 4 trigrams	**Between 5 and 8 trigrams**	**Between 9 and 12 trigrams**	**Between 13 and 16 trigrams**	**Between 17 and 20 trigrams**
Males	0	2	3	0	0
Females	1	5	8	1	0

Table 1

Calculate the percentage of participants who recalled between 9 and 12 trigrams correctly, out of all the participants in the experiment.

(2)

Percentage .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

4

P 7 0 4 8 6 A 0 4 4 8

4. Question 1 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

2 Kamira is investigating morality. She wants to find out how likely people are to return lost items. Kamira conducted an observation for her investigation.

Kamira went to a large shopping centre and positioned herself in the food court where people often sit to eat and have been known to leave personal items behind when leaving.

When someone had left an item behind, Kamira observed the behaviour of all passers-by who saw the item. She recorded whether they:

- looked at the item, but then ignored it
- picked up the item and gave it to a member of staff
- walked away with the item.

(a) Describe why Kamira may have used an observation research method for her investigation.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 2 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Draw a frequency table that Kamira could have used to record the data for her observation.

You must include appropriate column and row headings.

(3)

OLevels.com SCHOOL

(c) Describe **one** situational variable that Kamira may have considered when conducting her observation.

(2)

.....

.....

.....

.....

.....

.....

(Total for Question 2 = 7 marks)

5. Question 2 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

2 Jenna investigated whether stress affected performance on a memory test.

She recruited 10 participants from her university. Participants were asked to remember a list of 20 words. Fifteen minutes later, they were asked to write down as many words as they could remember. The same participants were tested twice:

- Test 1: one week before their examinations began.
- Test 2: one week after their examinations had finished.

(a) Describe how Jenna could use an opportunity sampling method to gather the participants for her investigation.

(2)

(b) State which experimental research design Jenna used in her investigation.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 2 · part b · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Jenna recorded the total number of correct words remembered by participants both before and after their examinations. Her results are shown in **Table 2**.

Participant	Total number of correct words remembered one week before examinations began (out of 20)	Total number of correct words remembered one week after examinations finished (out of 20)
A	5	8
B	6	7
C	7	9
D	4	8
E	5	7
F	3	7
G	9	9
H	10	11
I	6	6
J	5	8

Table 2

OLevels.COM SCHOOL

(c) Calculate the mean score for the total number of correct words remembered one week **before** the participants' examinations began.

(1)

Mean .....

(d) Calculate the median score for the total number of correct words remembered one week **after** the participants' examinations finished.

(2)

Median .....

(Total for Question 2 = 6 marks)

6. Question 2 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Experiments: lab, field and natural

Features, strengths and weaknesses of the three experimental methods.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-05

1. Question 5 · part whole · 12 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

***5** Godden and Baddeley (1975) conducted a field experiment to test memory. They used lists of words that were learned and then recalled in either the same or different environments to learning.

They used scuba divers as participants who learned and recalled the words either underwater or on dry land. These environments were familiar to the divers.

The scuba divers were allocated into the following different conditions:

- Words learned underwater, recalled underwater.
- Words learned underwater, recalled on dry land.
- Words learned on dry land, recalled on dry land.
- Words learned on dry land, recalled underwater.

The word lists were prerecorded and presented to the scuba divers twice when they were in their learning environment. The words were read at two-second intervals in the recording.

Godden and Baddeley (1975) found that more words were recalled when the divers were in the same environment in which they had learned the words, compared to when learning and recall took place in different environments.

Evaluate the use of field experiments to investigate human behaviour.

(12)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

1. Question 5 · part whole · 12 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

1. Question 5 · part whole · 12 marks

2023 JUN · Paper 02

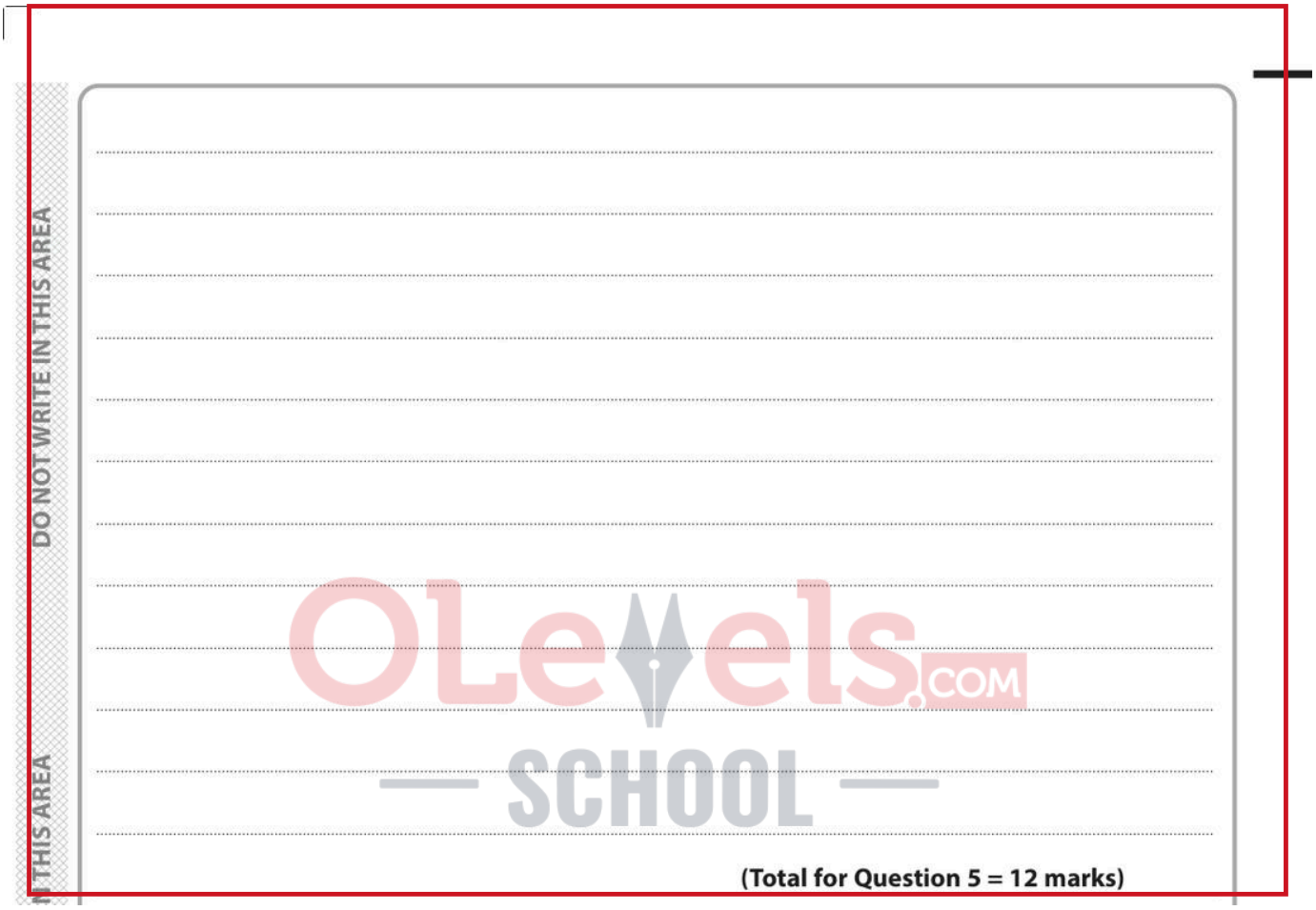
Worked-example retry · Attempt part whole only.



1. Question 5 · part whole · 12 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.



1. Question 5 · part whole · 12 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part a · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3 Becca wants to find out if there are sex differences in male and female brain functioning. She decides to use a laboratory experiment.

(a) State **one** feature of a 'laboratory experiment'.

(1)

(b) Describe how Becca could use volunteer sampling to gather the participants for her laboratory experiment.

(2)

OLevels.COM SCHOOL

(c) Becca will give each participant a coordination task and a word-meaning task to test for sex differences in brain functioning. She decides to use a single-blind technique.

Describe **one** reason why Becca used a single-blind technique for her experiment.

(2)

(d) Becca conducted her laboratory experiment with 30 male and 25 female participants.

Calculate the ratio of male to female participants in Becca's sample.

(1)

Ratio .....

2. Question 3 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA DO NOT WRITE

1 James is investigating the impact of brain damage on memory. He asks a brain-damaged patient to come to the university where he has set up a room to conduct a test of word recall.

(a) Identify the research method James is using in his investigation.

(1)

- ☒ **A** Questionnaire
- ☒ **B** Laboratory experiment
- ☒ **C** Natural experiment
- ☒ **D** Interview

(b) Describe how James can maintain the confidentiality of the patient when he publishes his investigation.

(2)

OLevels.COM SCHOOL

(Total for Question 1 = 3 marks)

Working and answer

4. Question 5 · part whole · 12 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

***5** Becker et al. (2002) conducted a natural experiment in Fiji to investigate the impact of television on eating behaviours. They assessed attitudes to eating in 1995, just as television was being introduced to the native population. They returned in 1998 to assess eating behaviours after three years of television exposure.

63 girls with a mean age of 17.3 years took part in 1995. A second sample of 65 girls with a mean age of 16.9 years took part in 1998.

To score eating behaviours, the two samples of girls completed a standardised EAT-26 questionnaire. They also answered questions about television viewing. Their weight, height, and body mass index (BMI) were also recorded.

Comparisons between 1995 and 1998 data indicated that, after exposure to western imagery on television, the reports of disordered eating such as self-induced vomiting to lose weight had increased.

Evaluate the use of natural experiments to investigate human behaviour.

(12)

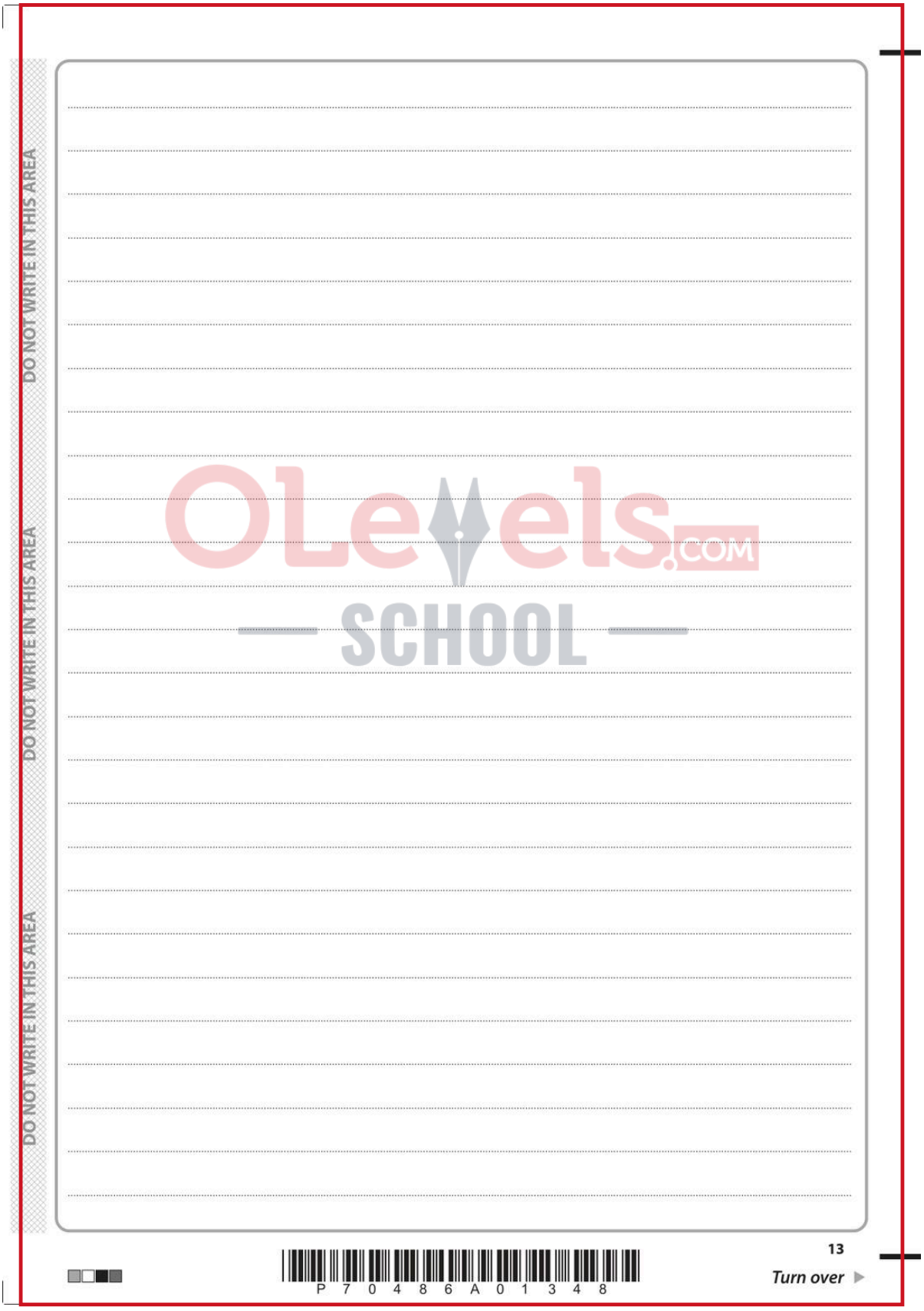
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 5 · part whole · 12 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.



4. Question 5 · part whole · 12 marks

2022 JUN · Paper 02

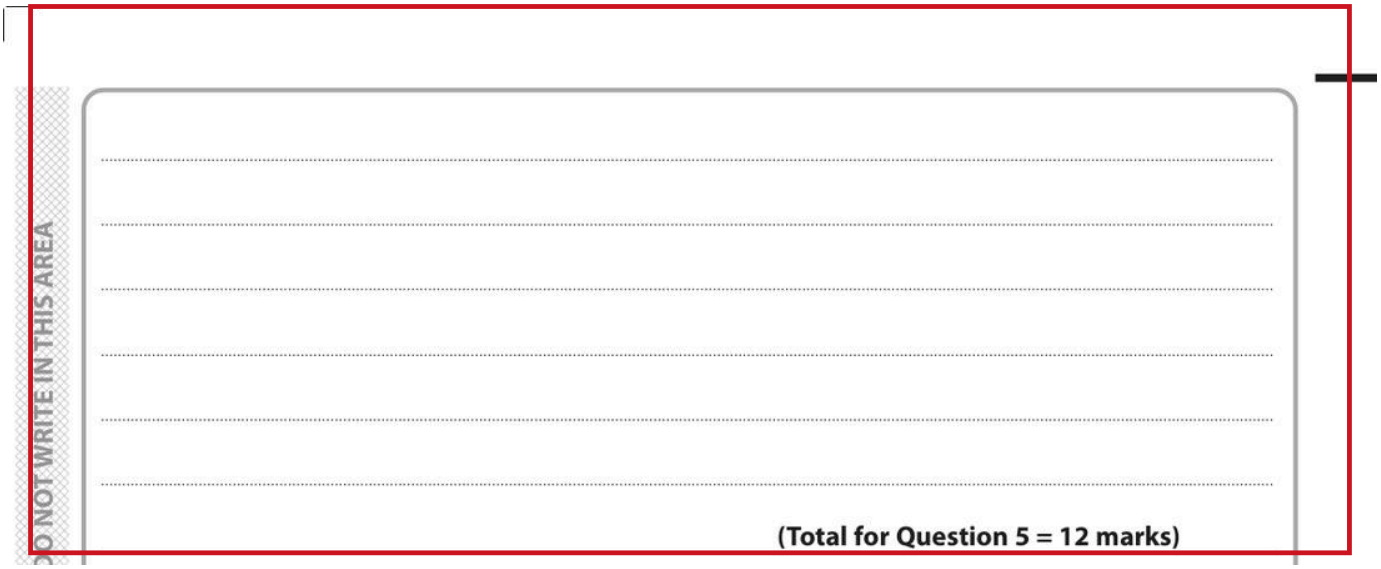
Further practice · Attempt part whole only.



4. Question 5 · part whole · 12 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.



OLevels.com SCHOOL

4. Question 5 · part whole · 12 marks · Working and answer

Working and answer

OLevels.com SCHOOL